Jhon Lito Padit

San Pedro, Laguna | +639050731321 | paditjhon@gmail.com | Portfolio: https://jjanan-ui.github.io/Portfolio/

PROFESSIONAL SUMMARY

Registered Mechanical Engineer with 3 years of combined experience in engineering, product development and manufacturing. Proven track record of leading cross-functional product engineering and New Product Introduction (NPI) projects from development through manufacturing. Expert in managing engineering changes, product specifications and technical documentation, including BOMs and drawing revision control. Strong ability to collaborate effectively with contract manufacturers, suppliers, and internal teams to resolve manufacturing issues and drive cost reduction.

CORE COMPETENCIES

Process: NPI, ECN, Design of Experiment, RCA and FMEA; Manufacturing Process and Machine Qualification; Product/Material Testing, Product/Material Validation and Qualification.

Drafting & CAD: SolidWorks 2D and 3D modeling; AutoCAD 2D and 3D modeling; sheet-metal flat patterns and bend allowances; Bill of Materials (BOM); drawing revision control; Design for Manufacturing (DFM); Laser Machine DXF file creation.

Engineering: Product Design Validation; Cost Reduction, Continuous Improvement, and Process Optimization; Technical Writing, SOPs, and Work Instructions; Supplier Coordination and Contract Manufacturing Collaboration.

Data Analysis: JMP, Excel, Power BI, SQL

WORK EXPERIENCE

Merlin Solar Technologies Inc. Laguna Technopark, Biñan, Laguna
Development Engineer II *Jun 2023 - Apr 2026*

- Lead and support product engineering and NPI projects from development through manufacturing, ensuring smooth transition of solar panel products from R&D to full-scale production across global sites.
- Manage engineering changes, product specifications, and document revisions by authoring and executing Engineering Change Notices (ECNs) to align with quality requirements and customer requirements.
- Work closely with external and contract manufacturers to support product development, manufacturing readiness, and technical changes, achieving >95% initial yield during factory startup in Cambodia and the Philippines.
- Coordinate product testing, validation, and qualification for high-performance materials, utilizing statistical analysis and Design of Experiments (DOE) to identify key process variables and enhance durability.
- Investigate and resolve product and manufacturing issues through root cause analysis (RCA) and Failure Mode and Effects Analysis (FMEA), reducing field failure rates and establishing new global material standards.
- Support cost reduction and continuous improvement initiatives by validating and implementing alternative materials, achieving a 30% cost reduction per watt and enhancing product competitiveness.
- Prepare technical reports, analyses, and project documentation to support product and manufacturing decisions, presenting data-driven recommendations to senior leadership.
- Monitor project timelines, risks, and deliverables to support successful product launches, ensuring performance metrics were consistently met or exceeded.
- Author 20+ SOPs and Work Instructions, driving process standardization and sustaining engineering efforts across global manufacturing sites.

Moog Controls Corporation Loakan Road, Baguio City
Intern as a Machinist *Dec 2016 - Apr 2017*

- Operated and set up CNC machines (lathe, milling) while performing precision measurements using micrometers and gauges.
- Gained practical exposure to mechanical tolerances, blueprint interpretation, GD&T symbols and machining workflows, informing accurate CAD drafting practices.
- Exposure to complex G-codes and controls of CNC machines.

EDUCATION

Technological Institute of the Philippines - Manila 363 Casal St, Quiapo, Manila
Bachelor of Science in Mechanical Engineering *Jun 2021*

Technological University of the Philippines - Manila San Marcelino St, Ayala Blvd, Ermita, Manila
Bachelor of Technology in Mechanical Engineering *Apr 2017*

PORTFOLIO & PROJECTS

GitHub Project Link: https://github.com/Jjanan-UI

1. Modular Bifold Door System

- Complete 3D CAD assembly of a space-saving bifold door system with pivots, hinges, brackets, and rollers; includes workshop drawings, BOMs, installation instructions, kinematic motion studies, and scalable 2/4-panel configurations with revision-controlled documentation.

2. Bifold Door Kit

- Documents the transition from 3D CAD to a shippable hardware kit, including a part catalog and part-numbered renders of custom components and fasteners for technical documentation and inventory management.

3. Custom Electronics Enclosure

- Designed a 2.0 mm Aluminum 5052 sheet-metal enclosure with mounting tabs, counterbores, cut-outs, and PCB mounting points; produced 2D flat patterns and fabrication drawings ready for laser cutting with DFM considerations.

4. Custom Electrical Enclosure

- Designed a 1.5 mm CRS steel enclosure with fan grills, louvered ventilation, and I/O cut-outs; produced a folded 3D model and 2D flat pattern applying bend allowances and manufacturing constraints.

5. Cellphone Holder

- Engineered a single-piece flat-pattern stand in 3.0 mm CRS with tool-less interlocking tabs, applying DFM to convert a 2D laser-cut profile into a 3D functional part.

LICENSES AND CERTIFICATION

Registered Mechanical Engineer: Philippine Professional Regulation Commission (PRC)

Lean Six Sigma White Belt: Innovation Philippines Training and Consultancy

BOSH Safety Officer 2: Department of Labor and Employment (DOLE)

Data Analyst Skill path: Zero to Hero in Excel, SQL, & Python - Udemy